Jamiul Kawsar

Computer Science & Engineering
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PROFILE

Computer Science and Engineering graduate with a strong academic background in **Computer Science, Artificial Intelligence, Machine Learning, and Pattern Recognition**, graduating with a **CGPA of 3.93/4.00** and receiving the **Vice-Chancellor Gold Medal**. Possess a solid foundation in programming, algorithms, data structures, machine learning, and software development, with hands-on experience developing projects using **Python, Scikit-learn, TensorFlow, and XGBoost**. Also experienced in embedded systems and autonomous robotics through an undergraduate thesis project. Passionate about teaching, mentoring students, and translating complex technical concepts into clear and practical learning experiences. Seeking an Instructor **position in Computer Science and Engineering** to contribute to teaching, student development, and academic activities.

EDUCATION

Sonargaon University | Dhaka, Bangladesh **2021-2025**

B.S. in Computer Science and Engineering | CGPA: 3.93/4.0

- **Relevant Courses:** Artificial Intelligence; Image Processing and Computer Vision; Pattern Recognition; Data Structures and Algorithms; Numerical Methods; Mathematics including Linear Algebra, Complex Variables and Laplace Transform; Calculus; Computer Networking; Data Communication; Database Management Systems; Software Engineering; Simulation Modelling;

ACADEMIC HONORS

- **Vice-Chancellor Gold Medal Recipient**
- **CGPA: 3.93/4.00**

UNDERGRADUATE THESIS

Design and Implementation of a Robot Car Based on ATmega328P

Designed and implemented an autonomous robot car with obstacle avoidance, Bluetooth control, and live video streaming, exploring autonomous navigation and embedded robotic-system design.
Technologies: Arduino, ATmega328P, Embedded C

AI / MACHINE LEARNING PROJECTS

Multi-Class Chest X-Ray Classification Using Deep Learning | ***PyTorch, CNN, ResNet50***

- Developed a reproducible deep-learning pipeline for **3-class chest X-ray classification** (**Normal, COVID-19, Pneumonia**) using **9,208 images**, including dataset auditing, controlled data splitting, preprocessing, and model evaluation.
- Implemented a **custom CNN baseline and ResNet50 transfer-learning models**, evaluating performance using precision, recall, F1-score, confusion matrices, and multiclass ROC-AUC; achieved **0.998 macro-average ROC-AUC** with fine-tuned ResNet50 on the validation set.

- Extending the study through **augmentation and class-weighting ablations, Grad-CAM explainability, error analysis, external validation, and robustness evaluation** to investigate model generalization and reliability

Technologies: Python, PyTorch, Torchvision, Scikit-learn, NumPy, Pandas, Matplotlib, CNNs, Transfer Learning, ResNet50, Medical Image Analysis

Calories Burnt Prediction

- Built regression models to predict calories burned from input features.
- Performed preprocessing, feature engineering, model comparison, and evaluation.
- Selected XGBoost based on comparative performance.

Technologies: Python, NumPy, Pandas, Matplotlib, Scikit-learn, XGBoost, Streamlit

Real Estate Web Application

- Developed a responsive real estate website to showcase properties with an intuitive and user-friendly interface.
- Implemented property listings and reusable UI components to improve browsing and navigation.

Technologies: React, JavaScript, Tailwind CSS, HTML, CSS, Netlify

PROFESSIONAL EXPERIENCE

Army War Game Centre | *Bangladesh Army* **June 2026 - August 2026**

- Developed Python applications for internal GIS workflows.
- Processed geospatial datasets using GeoPandas and Rasterio.
- Collaborated with engineers to improve software reliability.

Technologies: Python, NumPy, GeoPandas, Rasterio, GIS, PyVista

Machine Learning Internship | *Softsasi (Remote)* **May 2026 - August 2026**

- Implemented **Python** utility functions and strengthened programming fundamentals for **machine learning workflows**.
- Performed **data preprocessing**, **filtering**, **grouping**, and **exploratory analysis** using **NumPy** and **Pandas** on the ML dataset.
- Built and evaluated supervised classification model using **Scikit-learn**, applying train/test splitting and **model evaluation** techniques.

Technologies: Python, NumPy, Pandas, Matplotlib, Scikit-learn, Git/Github

TECHNICAL SKILLS

Languages (**by proficiency**)**:** Python, C/C++, Java, JavaScript
Machine Learning: Scikit-learn, XGBoost, NumPy, Pandas, Matplotlib, Data Preprocessing, Feature Engineering, Model Evaluation
Deep Learning: TensorFlow, PyTorch, ANN, CNN
Tools: Git/GitHub, Linux(Ubuntu), VS Code, Jupyter Notebook, Google Colab, Kaggle
Other: React, Vite, Tailwind CSS, MySQL

LANGUAGES

Bengali - Native | English - Professional | Hindi - Conversational